

Sabia Richards

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Professional Experience

Staff Machine Learning Engineer

AB InBev, July 2022 – September 2025

Technologies: Python, Tensorflow, Unit-Testing, CI/CD, C++

- Provided model speedups and memory usage reductions in excess of 100x.
- Led a team to migrate code to modern frameworks which we could further iterate on sustainably.
- Embedded mathematical guarantees of business principles in the model architecture of the model greatly improving statistical performance.
- Built a distributed framework to optimize marketing budgets across multiple axes and constraints with low latency.
- Helped develop image stitching and recognition software for use by field technicians using iOS devices.

Mobile CI/CD Engineer

ADT Security, August 2021 – April 2022

Technologies: Jenkins, Fastlane, iOS, Android, Gradle, XCode, AppCenter

- Updated and maintained a Jenkins instance
- Created and debugged Jenkins pipelines for a variety of projects for mobile devices.
- Helped design automated builds and releases gated by the Jenkins pipelines.
- Monitored error reports from app use in the wild.

Software Developer

Zemoso, December 2018 – December 2020

Technologies: Java, Swing, Python, Javascript, JMeter, Kubernetes, React-Native, AWS, Android

- Migrated and refactored an Android app to MVVM Architecture.
- Worked on a React Native cross-platform app and native bindings to third party integrations.
- Tested, analyze, and designed improvements to code and infrastructure so metrics surpassed business requirements.
- Managed, monitored and tested production deployments using the Elastic Kubernetes service on AWS.

Lead Software Developer

smartObjx, August 2017 – Dec 2018

Technologies: C#, Angular, VSTS, Unit-Testing, SQL, RavenDB, Agile

- Built and wrote a system for safe database migrations.
- Developed REST API microservices.
- Automated library generation to allow support for several languages with little to no long term overhead.

Mobile SDK Developer

Apptimize, December 2015 – July 2017

Technologies: A-B-Testing, Android, iOS, Java, Linux, Objective-C, Scrum, Unit-Testing, Continuous-Integration

- Built and deployed new features, third party integrations, and bug fixes for Apptimize's A/B and feature flag testing framework.
- Wrote end to end tests for continuous integration and associated testing procedures.

Speech Engineer

Bumpers, April 2015 – August 2015

Technologies: Android, C++11, Cross-Platform, iOS, Linux, Speech-Recognition

- Explored the possibility of using speech diarization in a cross-platform mobile app using the SHoUT library.
- Built a library using `kiss_fft` to separate out blocks of speech.
- Evaluated third party libraries on basis of required dependencies and licensing requirements.

Mobile Developer

Fill My Blank, August 2014 – February 2015

Technologies: Android, C++, Cross-Platform, iOS, Json, Libcurl, REST

- Developed and prepared to deploy a mobile application across multiple platforms.
- Focused on optimizing network traffic and on-device code.

Open Source Work

Magic the Gathering: Machine Learning (MtgML)

<https://github.com/CubeArtisan>

Technologies: Python, Tensorflow, Transformers, NLP, Recommendation

- Created several models for predicting the next pick that a player would make in a given Magic the Gathering draft in an arbitrary environment.
- Built models for recommendations on additions and removals to cubes (collections of cards used for drafts).
- Tested approaches for model to build decks given the cards to choose from, had some promising results, but could use more research.

DraftbotOptimization

<https://github.com/ruler501/DraftbotOptimization>

Technologies: C++, CUDA, SYCL, CMA-ES, Black-Box-Optimization

- The first iteration of what would become the MtgML models using black box optimization techniques.
- Used heterogeneous compute libraries to allow accelerating optimization on whatever compute hardware was available.

CubeArtisan

<https://github.com/CubeArtisan/cubeartisan>

Technologies: Typescript, NodeJS, React, SolidJS, WebSockets

- A website made for managing Magic the Gathering cubes (collections of cards used for drafting).
- Provided a fully-fledged domain specific filtering language.
- Using the filtering language allowed for creating complicated assignments of cards to packs while maintaining randomness.
- Allowed calculating custom analytics on the complicated random distributions of cards into packs.
- Supported real time and asynchronous multiplayer drafting.

CubeArtisan Infrastructure

<https://github.com/CubeArtisan/cubeartisan-infrastructure>

Technologies: Kubernetes, CI/CD

- Deployment scripts and configurations for provisioning a kubernetes cluster for CubeArtisan.
- Supported multiple different environments for production and staging.
- Managed deployment of the MtgML models to be consumed by CubeArtisan and other sites.

Mathematics Research Programs

<https://github.com/ruler501/Rationals>

Technologies: C++11, HPC, Math, Multithreading

- A collection of programs accomplishing a variety of tasks for a mathematics research group.
- Highly optimized and designed to run in parallel on a high performance computing cluster.
- Able to utilize heterogeneous compute hardware.

Probability Simulator

<https://github.com/ruler501/ProbSim>

Technologies: Probability, Statistics, C++17, Functional-Programming, Templates, Variadic-Function

- Developed a library for calculating discrete distributions and transformations on them.
- Supported both a lossless and mildly lossy mode that was much faster.
- Took functions applied to the outputs of a distribution to create new distributions.

Retrieve Green Autonomous Robot Software

<https://github.com/ruler501/retgreen>

Technologies: C++, OpenCV, PID-Controller, Robotics

- Wrote code to run on a robot designed by my team.
- Searched for and pick up poms of a specified color while avoiding approaching those of other colors.
- Utilized a PID system and custom real time computer vision to determine exact positions.

Corsair Keyboard and Mouse Drivers

<https://github.com/ckb-next/ckb-next>

Technologies: C, KissFFT, Linux, Drivers

- Built in new visual effects for the Corsair K65/K70/K95/Strafe Driver for Linux and OSX.
- Visualized music using a discrete fourier transform.
- Tracked a heatmap based on what keys were being used.